

Bitcoin & Quantum Computing

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Bitcoin & Quantum Computing

- ✧ Bitcoin threat model
- ✧ Current state
- ✧ Timeline for Solution



Quantum computing

- ✦ Leverages the principles of quantum mechanics
- ✦ Allows significant speed-ups in solving specific problems
- ✦ Cryptographically Relevant Quantum Computers (CRQCs) could break ECC-based cryptography
- ✦ To date, no quantum computer has outperformed a supercomputer on any commercially relevant application

CRQC Timeline Assessment

- ✦ US & UK - complete migration to PQC by 2035
- ✦ Several countries (including the US) target 2030 for significant progress
- ✦ Experts give >50% chance of a CRQC in the next decade
- ✦ Nvidia's CEO recently estimated it may take at least 20 years
- ✦ *"I think today that message needs to change. I think today the message needs to be: yes, unequivocally, worry about this now."* Scott Aaronson

Quantum computing and Bitcoin

- ✧ Mining
 - ✧ Quantum miners running Grover's Algorithm efficiently
- ✧ Signatures
 - ✧ CRQCs running Shor's Algorithm breaking cryptographic assumptions of ECC

Mining

Mining goal: Find a nonce that hashes below a target

Classical mining: More attempts = higher chance

- Linear growth, parallelisable

Quantum mining: Longer runtime = higher chance

- Square-root speedup, not parallelisable

Bitcoin Threat Model - Mining

- ✦ Mining will be significantly less affected in the foreseeable future
- ✦ If quantum mining will become realistic, we can expect:
 - ✦ Significant mining centralisation
 - ✦ Increase in stale blocks
 - ✦ Weakening of security assumptions (51% attack)

Asymmetric Cryptography

- ✦ There are two keys - Public Key (**PK**) and Secret Key (**SK**)
- ✦ A **SK** can be used to sign a message
- ✦ The base assumption are:
 - ✦ Calculating **PK** from **SK** - **Easy**
 - ✦ Calculating **SK** from **PK** - **Infeasible**
 - ✦ Forging a signature having only **PK** - **Infeasible**

What would CRQCs break?

- ✦ **EVERYTHING!** (Well, almost)
- ✦ Calculating **SK** from **PK** - **Easy**
- ✦ Forging a signature having only **PK** - **Easy**
- ✦ The attacker needs access to the **PK**
- ✦ Quantum secure signatures are known, but **lack efficiency or capabilities**
- ✦ There is **active ongoing research** in the cryptography community

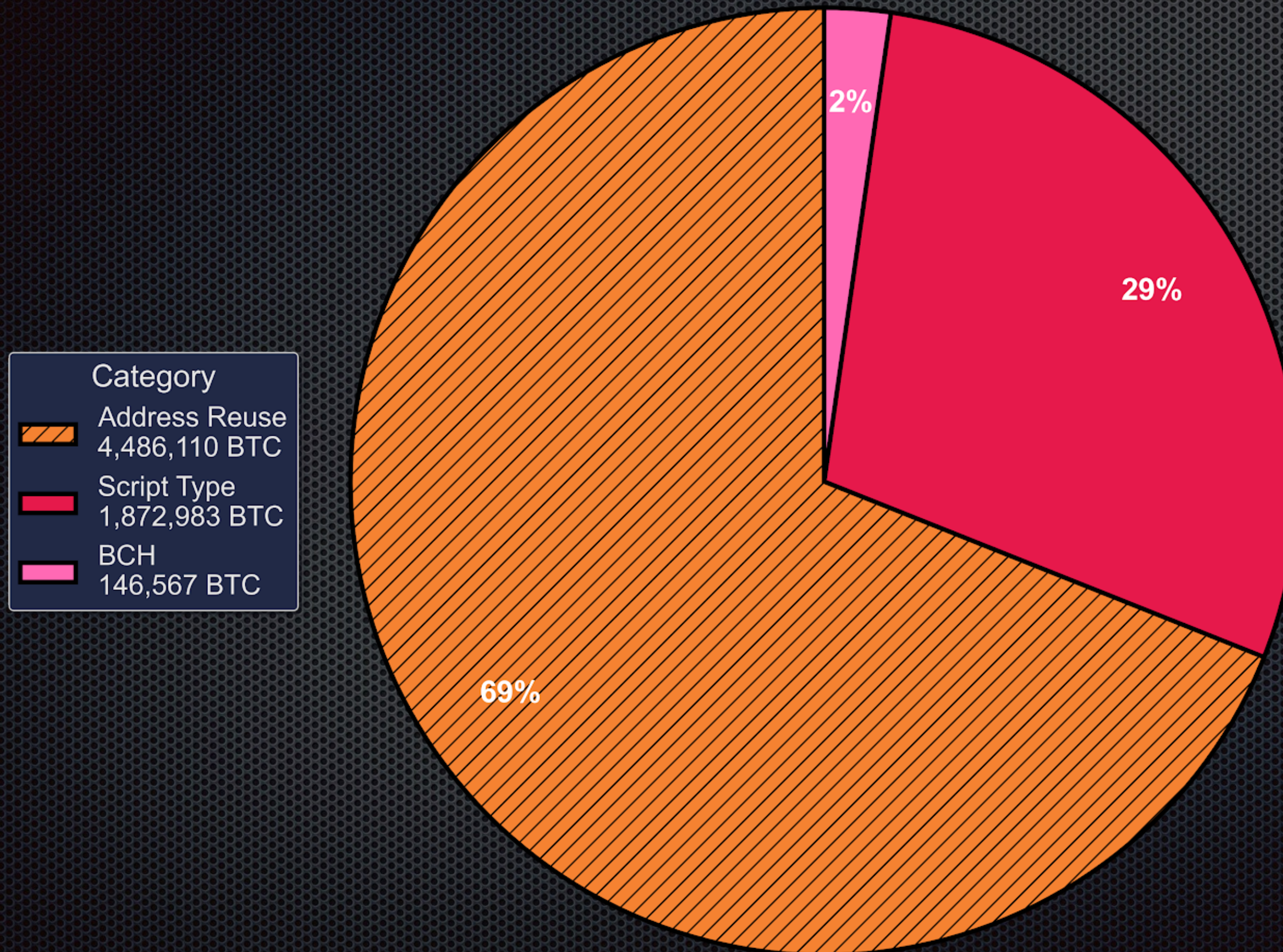
Cryptography and Bitcoin

- ✦ For a bitcoin transaction to be valid it needs to be signed by the **SK** matching the **PK** of the UTXO
- ✦ In some cases the **PK** is present on-chain
- ✦ In others a **hash** of the **PK** is on-chain, and needs to be revealed together with a signature to move the funds

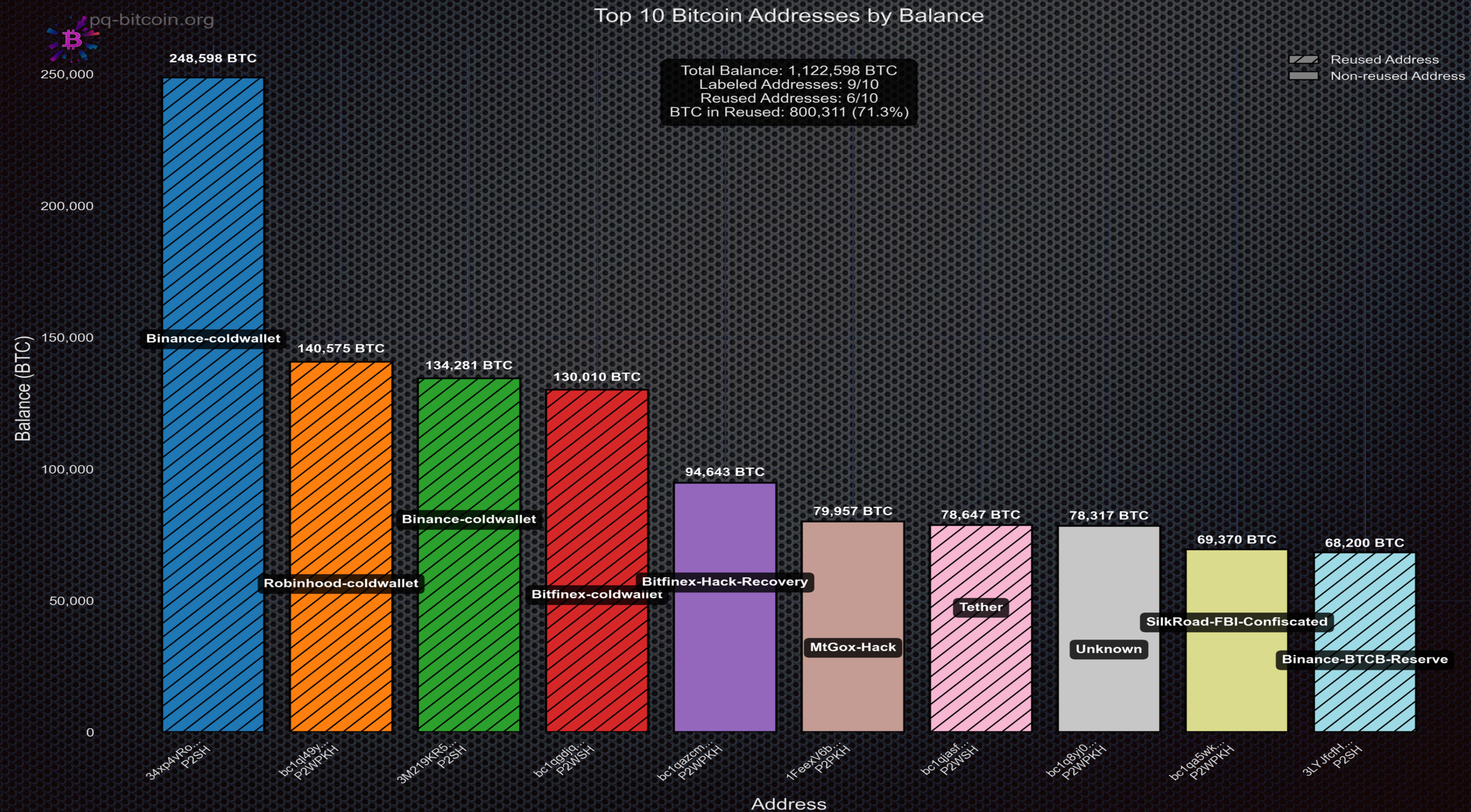
Bitcoin Threat Model - Signatures

- ✦ Some UTXOs are **immediately** vulnerable (long-range attack)
 - ✦ P2PK, P2MS, P2TR
 - ✦ Address re-use
 - ✦ BCH and other forks
- ✦ All other will be vulnerable **upon spend** (short-range attack)

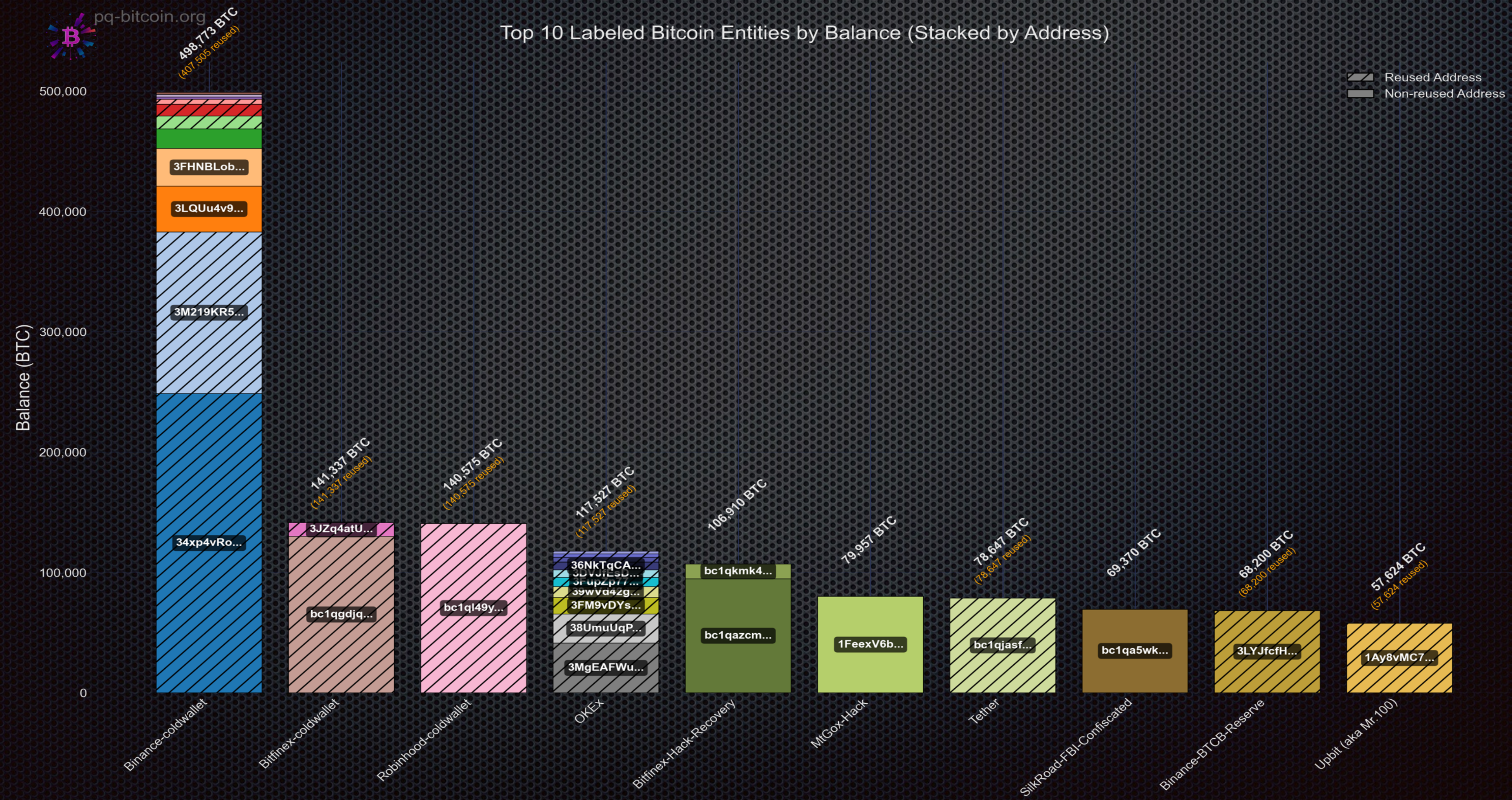
Immediately vulnerable ~6.51M BTC



Top Addresses



Top 10 labeled



Reused addresses

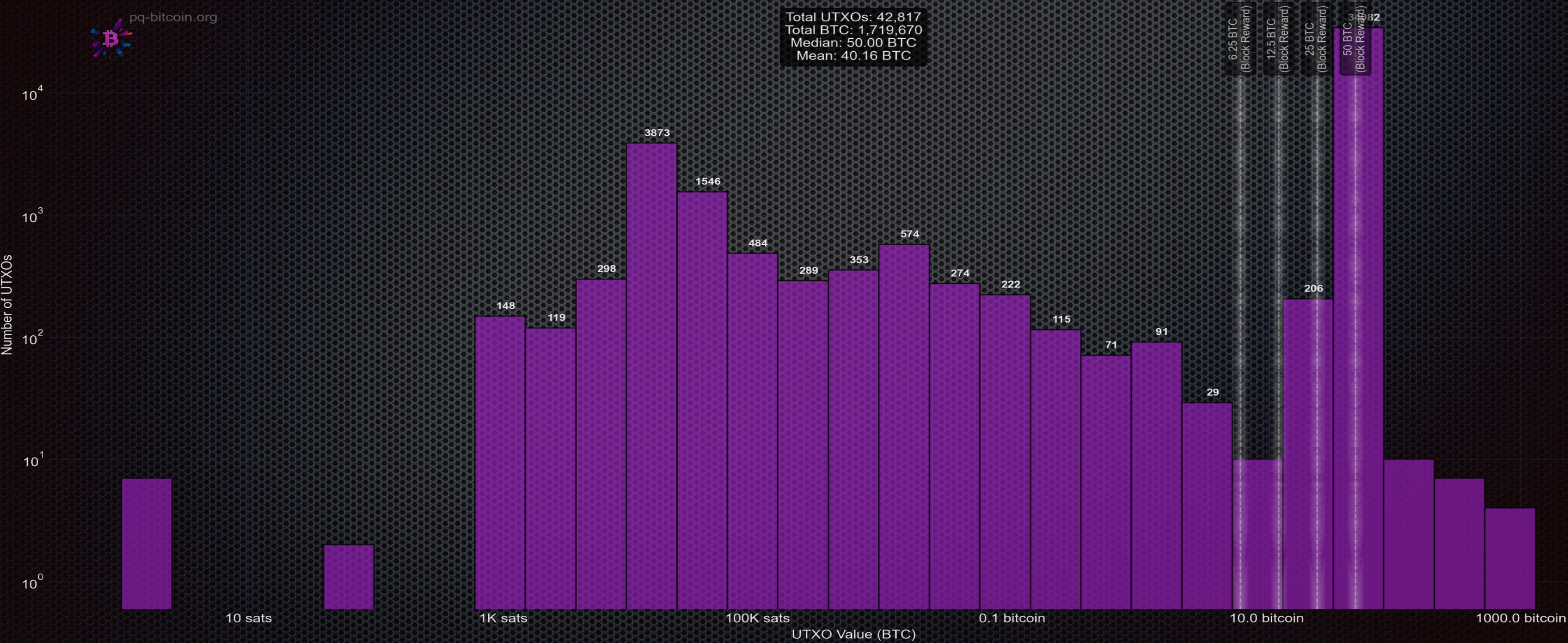
- ✦ Most of the funds have been active recently
- ✦ Significant amounts are with known entities

Activity Period	Addresses	Address %	Total BTC	BTC %
Within last 5 years	5354856	63.73	4371782.98	97.45
Between 5 and 10 years	2224323	26.47	64172.36	1.43
Between 10 and 15 years	823216	9.80	49784.88	1.11
More than 15 years	221	0.00	370.20	0.01

P2PK ~1.72M BTC

P2PK UTXO Value Distribution

Total UTXOs: 42,817
Total BTC: 1,719,670
Median: 50.00 BTC
Mean: 40.16 BTC



QV The story so far

- ✦ ~6.51M BTC (32.73% of total supply) is immediately QV
- ✦ Address Reuse ~4.49M BTC likely can be moved
 - ✦ T1000: account for ~2.73M BTC high rates of reuse 30% (vs. 15-20%)
- ✦ P2PK 1.72M BTC almost entirely coinbase TX, likely can't move
- ✦ P2TR 0.15M BTC many non-economic UTXOs

Script & address risk

Script type	Example	Vulnerability
P2PKH	1A1zP1eP5QGefi2DMPTfTL5SLmv7DivfNa	Hashed - OK if not reused
P2SH	3J98t1WpEZ73CNmQviecrnyiWrnqRhWNLy	Hashed - OK if not reused
P2WPKH	bc1qw508d6qejxtdg4y5r3zarvary0c5xw7kv8f3t4 (bc1q + 38 more characters)	Hashed - OK if not reused
P2WSH	bc1qrp33g0q5c5txsp9arysrx4k6zdkfs4nce4xj0gdcccefvpysxf3qc cfmv3 (bc1q + 58 more characters)	Hashed - OK if not reused
P2TR	bc1p5d7rjq7g6rdk2yhzks9smlaqtedr4dekq08ge8ztwac72sfr9rusx g3297 (bc1p + 58 more characters)	Inherently vulnerable

Immediate Protective Measures

- ✦ **DO NOT REUSE ADDRESSES!**
- ✦ Large amounts held in vulnerable addresses can be moved
 - ✦ P2PKH, P2SH, P2WPKH, or P2WSH
- ✦ For old coins - check if addresses reuse occurred on forks
- ✦ Don't hold funds in exchanges (or verify how they manage key security)

Migration Pathways

- ✦ Migration of all utxo can take between 3 months and 2 years or more
- ✦ There are several mechanisms, some permissive and some protective:
 - ✦ CDR - Commit, Delay, Reveal
 - ✦ Hourglass strategy - Rate limit vulnerable UTXO spend
 - ✦ QRAMP - Enforcing mandatory migration period

Burn vs. Steal Dilemma

- ✦ Treat **CRQCs** as a “Bug”
- ✦ What do we do with coins that did **not migrate**?
- ✦ **Burn** - coins can never move
 - ✦ Philosophical clash
- ✦ **Steal** - Assets claimed by those with a CRQC
 - ✦ Destabilise the ecosystem
- ✦ **Hourglass** - rate limit

Timeline for upgrade

Three main phases:

1. Research

2. Implementation

3. Migration

Long-Term

2.5 Y

1.5 Y

3 Y

~7 Y

Contingency

0.5 Y

1 Y

1 Y

~2 Y

Bitcoin & Quantum Computing

- ✦ Bitcoin threat model
 - ✦ Mining
 - ✦ Signatures
- ✦ Current state of the UTXOs
- ✦ Timeline to solution



For more details:

pq-bitcoin.org



PQ report:

[chaincode.com/bitcoin-
post-quantum.pdf](https://chaincode.com/bitcoin-post-quantum.pdf)

